

# Next Generation Personalised Medicine for all (NG-PM4all)

## 1. General information

### 1.a Applicants

| Name                                      | Institution  | Department                                                                                     | Email                         | Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify) | For Erasmus MC applicants: clinician/ researcher/ mix | Role in Flagship [Main Lead/Lead/ Applicant] |
|-------------------------------------------|--------------|------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|
| <b>Prof.dr. Hester Lingsma</b>            | <b>EMC</b>   | <b>Public Health</b>                                                                           | <b>h.lingsma@erasmusmc.nl</b> | <b>T</b>                                                            | <b>R</b>                                              | <b>Main lead</b>                             |
| <b>Prof.dr. Maureen Rutten-van Mölken</b> | <b>EUR</b>   | <b>Erasmus School of Health Policy &amp; Management</b>                                        | <b>m.rutten@eshpm.eur.nl</b>  | <b>T</b>                                                            | <b>R</b>                                              | <b>Lead</b>                                  |
| <b>Prof.dr.ir. Caspar Chorus</b>          | <b>TUD</b>   | <b>Engineering Systems and Services</b>                                                        | <b>c.g.chorus@tudelft.nl</b>  | <b>T</b>                                                            | <b>R</b>                                              | <b>Lead</b>                                  |
| Dr. David van Klaveren                    | EMC          | Public Health                                                                                  | d.vanklaveren@erasmusmc.nl    | T                                                                   | R                                                     | Appl                                         |
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| Dr. Maryam Kavousi                        | EMC          | Epidemiology                                                                                   | m.kavousi@erasmusmc.nl        | T                                                                   | R                                                     | Appl                                         |
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| Prof.dr. André Uitterlinden               | EMC          | Internal Medicine                                                                              | a.g.uitterlinden@erasmusmc.nl | T                                                                   | R                                                     | Appl                                         |
| Dr. Jeroen van Rooij                      | EMC          | Internal Medicine                                                                              | j.vanrooij@erasmusmc.nl       | TT                                                                  | R                                                     | Appl                                         |
| Dr. Iris Lansdorp-Vogelaar                | EMC          | Public Health                                                                                  | i.vogelaar@erasmusmc.nl       | T                                                                   | R                                                     | Appl                                         |
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| Prof.dr. Bruno Stricker                   | EMC          | Clinical Epidemiology                                                                          | b.stricker@erasmusmc.nl       | T                                                                   | C and R                                               | Appl                                         |
| Prof.dr. Liesbeth van Rossum              | EMC          | Internal Medicine                                                                              | e.vanrossum@erasmusmc.nl      | T                                                                   | C and R                                               | Appl                                         |
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| Prof.dr.. Anne Stiggelbout                | EUR/<br>LUMC | Erasmus School of Health Policy & Management                                                   | a.m.stiggelbout@lumc.nl       | T                                                                   | R                                                     | Appl                                         |
| Prof.dr. Dennis Fok                       | EUR          | Erasmus School of Economics                                                                    | dfok@ese.eur.nl               | T                                                                   | R                                                     | Appl                                         |
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| Dr mr. Niek Mouter                        | TUD          | Engineering Systems and Services                                                               | N.Mouter@tudelft.nl           | T                                                                   | R                                                     | Appl                                         |
| José Ignacio Hernández                    | TUD          | Engineering Systems and Services                                                               | j.hernandez@tudelft.nl        | O (PhD)                                                             | R                                                     | Appl                                         |
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*Planned start date of the seed funding:* 31-12-2022

*Primary convergence theme:* Improving health journeys

*Keywords:* Personalised medicine, genetics, prevention, treatment, data-analytics, patient-involvement, cost-effectiveness

### 1.b Summary (300/300 words)

**Personalised Medicine (PM) is a promising approach to improve individual health journeys,** while maintaining affordability of the health care system, both **central in the Health and Technology Convergence mission**. Instead of a “one-size-fits-all” approach, Personalised Medicine (PM) uses individuals’ phenotypes and genotypes for tailoring therapeutic strategies, to determine predisposition to disease, and to deliver timely and targeted prevention.

**In ‘Next Generation Personalised Medicine for all’ (NG-PM4all) we integrate data sources, perspectives, and approaches to develop personalised prevention and treatment strategies, to evaluate their impact on individuals and society, and to successfully embed them in routine practice.**

We recognize the **interdependencies between technology, people, and their organizational context**, using a **mixed-method approach** with quantitative and qualitative methodologies. We initially work on 6 clinical case-studies from **three major areas: cardiovascular disease, cancer, and mental health**, to create direct impact.

Work package (WP) 1 connects readily available phenotypic data, genetic data, and patient preferences to develop personalised treatment models with advanced data-analytics. These feed into WP2 which evaluates the (cost-)effectiveness and societal value of PM. WP3 implements the models from WP1 in trustworthy decision support tools. WP4 guides implementation of effective PM in public policies and clinical practice in a cost-effective, equitable and accessible manner. WP5 embeds values and preferences of citizens, patients and clinicians in the design of PM, and integrates the decision support tools from WP3 in patients’ health journeys. Values and preferences will also feed back into the other WPs to be considered in the development, evaluation and implementation of PM approaches. WP6 combines the methodological results of each WP in a **novel framework for PM** and creates roadmaps to PM for each case-study. We establish an **interdisciplinary expert centre on next-generation PM** that combines all our expertise to create **scientific, societal and clinical impact**, through personalising prevention and treatment.

### 1.c Public Summary (299/300 words)

**Het personaliseren van preventie en behandelingen is een veelbelovende manier om de gezondheid van mensen tijdens de levensloop te verbeteren** en tegelijk de zorg betaalbaar te houden. In plaats van een "uniforme" benadering, gebruikt “personalised medicine (gepersonaliseerde gezondheidszorg)” (PM) kenmerken van het individu, zoals gezondheidskenmerken en genetische informatie, om zorg op maat te leveren.

In dit voorstel **integreren we verschillende gegevensbronnen, perspectieven en benaderingen om PM te ontwikkelen en verbeteren, de impact daarvan op individuen en de samenleving te evalueren en bij te dragen aan goede inbedding van PM in de zorg.**

We kijken verder dan data en technologie om **recht te doen aan menselijke aspecten en context**. Daarvoor gebruiken we kwantitatieve en kwalitatieve methoden.

We werken in eerste instantie aan 6 ‘casussen in **drie grote ziektegebieden: hart- en vaatziekten, kanker en geestelijke gezondheid**.

Werkpakket (WP) 1 analyseert data uit verschillende bronnen, om gepersonaliseerde behandelmodellen te ontwikkelen. Deze worden geëvalueerd in WP2, om hun (kosten)effectiviteit en maatschappelijke waarde vast te stellen. WP3 implementeert de modellen uit WP1 in transparante en betrouwbare keuzehulpen voor patiënten en zorgverleners. WP4 onderzoekt hoe een kosteneffectieve, rechtvaardige en toegankelijke implementatie van PM in beleid en praktijk gerealiseerd kan worden. WP5 brengt de waarden en voorkeuren van burgers, patiënten en zorgprofessionals in kaart, om die te gebruiken bij het ontwikkelen van PM. Waarden en voorkeuren die daarbij naar voren komen, worden teruggekoppeld naar alle andere werkpakketten om deze mee te nemen in de ontwikkeling, evaluatie en implementatie van PM. WP6 combineert de wetenschappelijke resultaten van elk WP in een nieuw raamwerk dat de basis vormt voor **‘routekaarten voor PM’** voor elke case-study. We starten een **interdisciplinair expertisecentrum** ‘Next Generation Personalised Medicine for all’ dat al onze expertise combineert en creëren daarmee **directe impact op wetenschap, maatschappij en zorg**, door preventie en behandeling verder te personaliseren.

## 2. Relevance to Convergence Health & Technology mission (2500/2500 words)

### 2.a Contribution to H&T mission and fit with H&T theme(s)

One of the most promising ways of **improving individual health journeys** is to personalize prevention and treatment throughout the course of peoples’ lives. This is referred to as **Personalised Medicine (PM)**. It is defined by the European Council Conclusion on PM for patients (2015/C 421/03) as *“a medical model using characterization of individuals’ phenotypes and genotypes [...] for tailoring the right therapeutic strategy for the right person at the right time, and/or to determine the predisposition to disease and/or to deliver timely and targeted prevention.”*

Our Flagship **NG-PM4all** focuses on high impact applications of personalising prevention and treatment with the short-term goal to make an impact on three exemplar disease areas, i.e., cardiovascular disease, cancer, and mental health. However, there are many examples of research into personalised medicine beyond these disease areas. Not only do we plan to broaden the scope to other areas in later phases, but also do we develop methods, models and tools that can already be transferred to these areas.

NG-PM4all contributes substantially to life-long health and well-being, from pre-birth to end of life (**mission 1 of H&T**) by developing, assessing, and implementing **PM at various stages of the entire health journey throughout life**. The selected PM interventions that we start with in this Flagship are designed towards having a major impact when translated and implemented into clinical application and public policies. The selection is such that they i) **span the entire health journey** from health promotion, prevention, early disease detection, to personalized treatment[1], and ii) **cross the borders of different sectors** of the healthcare system by combining data from hospitals, primary care, and screening programs. In addition, they are in different phases of development and implementation of PM. All interventions involve (risk) prediction, which we base on **multi-layered data sources** that combine data on clinical characteristics (e.g., blood tests, physiological tests, imaging), genetics, lifestyle, environmental, socioeconomic, and self-reported characteristics, such as personal values and preferences[2].

In the **treatment domain** of our proposal, we increase access to individual-tailored precise medical treatment for all (**mission 2 of H&T**). We will stimulate **pharmacogenetic profiling** and evidence-based dosing of **cardiovascular, oncological and antidepressant drugs** using a pharmacogenetic profile app. This app supports the application of knowledge about genetic mutations in genes coding for enzymes involved in drug-metabolism. These mutations cause individual variation in response to drugs and susceptibility to side-effects/adverse events. As the proportion of patients with an actionable genotype is as high as 90%, knowing their pharmacogenetic profile before treatment-initiation will improve their health journey at any point in life where they need these drugs. Furthermore, in cancer, our strategies include the **personalisation of immunotherapy** (demonstrated in melanoma) and biomarker-guided therapies that do not depend on tumour-type, but on the presence of a genetic marker in the tumour regardless of the tumour's tissue of origin (histology-independent therapies). Immunotherapy is effective in improving survival but is expensive and associated with side-effects. Over the years, the approved indications for immunotherapy in melanoma have expanded and many patients are overtreated and experience side-effects. We develop, validate, and implement risk prediction models for the progression of various stages of melanoma to shorten treatment duration and guide safe stopping of immunotherapy. **Histology-independent therapies** personalize cancer treatment by tailoring them to (often rare) biomarkers in tumour tissue. The first histology-independent treatments were approved recently and many more will soon reach the market. These therapies pose great challenges for Health Technology Assessment (HTA) bodies advising on reimbursement, as they are often approved based on immature evidence from small, single-arm basket trials.

In the **preventive domain** of our proposal (**mission 3 of H&T**), we combine polygenic risk scores and CT scan data with conventional risk prediction tools for **coronary artery disease** to refine individual risk estimates and offer cost-effective risk-based **lifestyle interventions**, not only to adults but also to younger individuals. We also advance **personalization of lung cancer screening** by combining (blood-based) biomarkers, CT scan results and polygenic risk scores with smoking and other phenotypical risk-factors, to stimulate the implementation of cost-effective population-based lung cancer screening programs. In the mental health domain, we not only focus on the pharmacogenetics of anti-depressants, but also on **attention deficit hyperactivity disorder (ADHD)**, which is the most common neurodevelopmental disorder in childhood and has a strong heritability factor as well as strong links to prenatal/perinatal factors. In ADHD we currently lack tools for early detection, and risk stratification is symptom-based. Polygenic risk scores or environmental risk factors alone lack predictive power and specificity. Therefore, we develop better risk detection tools for ADHD based on multimodal and developmentally sensitive markers.

Compared with less effective one-size-fits-all approaches, PM, by definition, increases some forms of inequality in healthcare. We address potentially **undesirable effects of inequality on equity (mission 4 of H&T)** by investigating:

- The representation of vulnerable groups in the databases we use;
- The correlation between biomarkers and indicators used for personalisation and equity-relevant variables, such as ethnicity, socioeconomic status, and health-literacy;
- The impact that PM might have on complicating reimbursement decision-making for small subgroups that result from more stratification that is inherent to PM (the smaller the groups become, the greater the uncertainty on outcomes);

- Concerns that the collection of personal data might lead to discrimination in relation to purchasing insurance or a mortgage;
- Means to address the specific needs of more vulnerable groups, so that they will benefit from improved access to PM;
- Means to develop **decision-making tools** that are conscious of equity and other ethical issues and facilitate patients' understanding of their treatment options and empower them in engaging in (joint) medical decision making.

Besides these equity issues PM comes with many other **challenges**. Although PM can be seen as a natural evolution of medicine, it is unique in the range and extent of challenges it faces. For example, regardless of the type of data used, PM requires some form of testing (infrastructure) that comes with costs. These costs must be accounted for in cost-effectiveness analyses of PM. As a result, a new PM intervention that is very beneficial for a small group of patients from an individual's perspective, might not be cost-effective from a societal perspective when it requires large-scale screening which exposes healthcare payers to high costs[3]. Hence, we need to assess the net benefits of PM from a societal perspective. Assessing net benefits is more complicated for PM than for non-PM because the uncertainty around the clinical effectiveness and cost-effectiveness may be higher as the subgroups of patients targeted become smaller. It is often impossible to perform an RCT for every subgroup, so we need to rely more strongly on alternative study designs, novel analytical methods and observational data. Appropriately addressing the heterogeneity in these data and determining the causal relationship between PM intervention and outcome is challenging. Furthermore, PM might generate outcomes beyond those that we commonly include in our cost-effectiveness analysis or outcomes beyond the current patient (e.g., conditions that result from a germline DNA variation and affect relatives).

NG-PM4all addresses these and other challenges and **advances the methodology** of developing, assessing, and implementing PM by:

- Improving (risk) prediction in high-dimensional data, analyse heterogeneity in intervention effects, and enhance personalised causal inference in non-randomized data to strengthen the evidence-basis on PM (WP1);
- Conducting full societal HTA's of test-intervention combinations that not only model clinical- and cost-effectiveness but also address organisational, social, ethical, and legal aspects of PM[4], to inform reimbursement decision-making (WP2);
- Designing transparent and easy-to-use decision support tools that aid both health professionals and patients in morally sensitive decision contexts to facilitate responsible use of PM interventions (WP3);
- Optimizing pricing- and reimbursement models for PM, planning (test-)infrastructure/ capacity, and embedding PM in care pathways to ensure early and wide access to PM (WP4);
- Embedding values and preferences of citizens, patients, and clinicians in the design of PM to ensure societal support and a smooth implementation (WP5).
- Integrating output from the WPs in a novel conceptual framework for PM, addressing all steps from development of PM approaches to evaluation to implementation (WP6).

## 2.b Synergy between disciplines

NG-PM4all merges expertise from many applicants currently working on personalised prevention and/or treatment or on analytical methodology that can directly be applied to PM, to ultimately result in a **centre of expertise in next generation PM**. Historically, they were working largely in silos, but in recent years many applicants have built a strong **track record in generating synergy across disciplinary boundaries**.

NG-PM4all includes **(clinical) experts on the PM interventions in cardiovascular disease, cancer, and mental health**. For personalised prevention, this pertains to experts in public health, population genetics, cardiovascular disease, population screening for cancer, and (child and adolescent) mental health. For personalised treatment, this pertains to experts in oncology (specifically melanoma), cardiometabolic medicine, pharmacogenetics, and laboratory science.

These experts will partner with experts in the field of **advanced data-analytics** (causal inference, modelling high-dimensional data, robust methods to estimate heterogeneous treatment effects), epidemiology (prediction and prognostic modelling, (observational) study design) and **full societal HTA** (cost-effectiveness modelling[5], HTA, pricing, reimbursement, ethical, legal, social, organizational aspects) to enhance the evaluation of PM and provide more robust evidence to optimise clinical and reimbursement decision making. To accelerate implementation and improve access to affordable and cost-effective PM interventions, experts in **pricing and payment models, organisational science, and ethical and legal aspects** are involved, as well as experts in **co-designing and implementing decision support tools** for patients and physicians. To investigate societal support for PM and ensure that it aligns with the values and preferences of citizens, patients, and professionals, experts in **preference measurement and citizen involvement** participate. While in isolation each of these experts would be able to make a marginal contribution to PM from their own discipline, together NG-PM4all can create the step-change which is needed to push the state of the art ahead and inspire other scientists and stakeholders.

## 2.c Long term sustainability plan

As PM has the wind in the sails, NG-PM4all is timely and adequate to accelerate and support the revolutionary transformation from a group-based to an individual approach in prevention and healthcare. NG-PM4all results provide a **solid basis for future expansion** of NG-PM4all into other disease areas and PM approaches.

This **expansion is feasible** as the consortium includes the required expertise, has access to a unique combination of population-based and patient-level datasets (Section 3), and is supported by patients, citizens, and other stakeholders of PM. Expansion is accelerated through many different community building efforts, the involvement of students and early career clinicians and researchers, and joint international grant applications (Section 4). The feasibility of this is illustrated by their excellent track record in obtaining funding, the scope of their (inter)national network, their strong roots in the clinic and in public health, their links to policy makers, and the alignment of NG-PM4all with (inter)national research agendas (Section 3d).

**Table 1: Mitigating risks**

| <i>Risk</i>                                                                                                                                                                                                                                                                                 | <i>Mitigation</i>                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethical, legal and privacy concerns about genetic profiling.                                                                                                                                                                                                                                | We invite/welcome these arguments, address the concerns, and build in safeguards in the PM approaches. Use GDPR-complaint data collection and ensure that procedures are inclusive.                                                                                                                                             |
| Difficulties engaging patients and citizens.                                                                                                                                                                                                                                                | We have involved patients and non-expert citizens in reviewing this proposal. Some had already contributed to previous research. To ensure that the right questions are being addressed and that the needs and concerns from their perspective are duly considered, we continue to involve them throughout NG-PM4all.           |
| Getting access to data.                                                                                                                                                                                                                                                                     | We will use existing databases that include a variety of data (e.g., phenotypic, genetic, clinical, imaging, socioeconomic, environmental, lifestyle and preference data). Permissions to use the data for research purposes have already been obtained. We also use clinical trial data of completed or about-to-start trials. |
| PM is beneficial from the perspective of the individual who gets the intervention but may not be so from a societal perspective, for example because many people must be tested to identify the few that can benefit from the intervention or because the intervention-costs are very high. | We will use cost-effectiveness models to estimate the parameter values (e.g., cut-off value of risk score, test-characteristics, treatment costs) at which the PM intervention will be cost-effective to steer the implementation of value-based PM.                                                                            |

## 2.d Visibility, attracting talents and new partners

Visibility activities to attract new talents and partners, overseen by the **Communication and Dissemination Committee** and the **Education Committee** (section 4d):

- **Policy makers, regulators, HTA-bodies, payers:** applicants are in continuous dialogue with high-level decision makers to align research questions/methods to their urgent needs;
- **Clinicians/patient organisations:** applicants are also in continuous dialogue with clinicians/practitioners/patient organisations outside NG-PM4all to safeguard that our research results and recommendations are of direct relevance to daily practice;
- **Academic community:** 1) top-level publications in open access journals, 2) presentations at conferences/symposia, 3) a dedicated website disclosing models, tools, and recommendations to guide the personalisation of interventions;
- **General public:** 1) social media, 2) animation films, 3) mass media (e.g., talk shows on national television/radio, popular science articles in (inter)national newspapers, science cafés)

- **Students:** 1) Minor program that brings together students from the three institutions and offers courses that integrate health-, data-, behavioural-, technical, and policy science, to train students in developing next-generation PM and translating their findings to public policies and clinical practice, 2) BSc and MSc-graduates participating in research projects of NG-PM4all, 3) group projects for BSc and MSc students on real-world PM cases;
- **Talents:** 1) establish a centre that functions as a contact address for talented (early career) researchers seeking interdisciplinary collaborations in PM, 2) regular seminars and workshops at each of the participating institutes, 3) continuously explore opportunities for new grant applications with members in our network (e.g., ICPeMed, 1+Million Genomes initiative).

## 2.e Long term societal impact

NG-PM4all catalyses the paradigm shift from a one-size-fits-all to a personalised approach by conducting fundamental research that addressed the challenges for large scale implementation of PM. The **impact pathway** is shown in Figure 2 (Section 3c).

### *Scientific impact*

NG-PM4all **advances methodology** in data-analytics, full HTA of broad societal benefits, person-centred decision support tools, innovative funding and payment models, conditions for successful embedding of PM in the system, and means of measuring societal support and being inclusive in engaging citizens and patients. Methods, models, and tools that are beneficial to other Flagships will be made available through easy to use open-source applications with rich web sources for documentation.

### *Health/wellbeing impact*

Current and future case studies will directly contribute to **more effective personalised prevention and treatment** strategies, not only for various cancers, cardiovascular diseases, and mental health problems but for many other disease areas as well. With a Minor program and training courses for professionals we will further **contribute to the implementation** of PM in clinical practice.

### *Societal impact*

By generating evidence for the cost-effectiveness and full societal value of PM, NG-PM4all ensures the implementation of effective PM interventions that are also **cost-effective for society**. NG-PM4all aims to improve **broader societal outcomes** like increased productivity, reduced healthcare- and informal care costs, and improved efficiency of the public health and healthcare system. All WPs in NG-PM4all will contribute to reduction in health inequalities on a societal level.

### 3. Flagship proposal (1995/2000 words excluding pre-specified headings)

#### 3.a Ambitions

Despite technical advancements in PM such as genetic and genomic profiling and availability of increasing amounts of data from various sources, the adoption, implementation, and wide-scale deployment of PM lag behind. **There are many challenges to be addressed** including improving data-analytics, providing robust evidence on societal value of PM, developing transparent and user-friendly decision support tools for responsible use of PM, whole systems analysis to improve embedding of PM in routine practice and increasing the societal support base for PM. **The ambition of NG-PM4all is to bridge the gap between all complementary expertise required to address these challenges, to ultimately establish an expert centre in next generation PM.**

#### 3.b Aims and feasibility

##### *Aim*

To integrate several data sources, perspectives, and approaches to develop and improve PM interventions, to evaluate their impact on individuals and society, and to develop solutions and strategies to speed-up the successful embedding of PM into routine practice.

##### *Feasibility*

We appoint **11 PhD students** to ensure execution of the tasks and output of the WPs. **Two postdocs** will be appointed for integration of all results. **We have ~900€k in-cash to match all Convergence funding in year 1 and 2** (Table 2 and Annex Budget). We expect to match at least 1.5 PhDs, 1 postdoc and material (800€k) in year 3-5 from proposals in the pipeline. **In total the consortium has at least ~€2.5M ongoing funding on the topic of PM and nine grants for at least ~€4.5M in the pipeline.** All grants considered for matching have (local) PI's participating in NG- PM4all and are in the project administration of the participating institutes. For future funding we will submit joint grant proposals. Concrete opportunities include the 'Dutch National Research Agenda', a call on innovation in early cancer detection and diagnosis from the Dutch Cancer Society, and the ZonMW Prevention program.

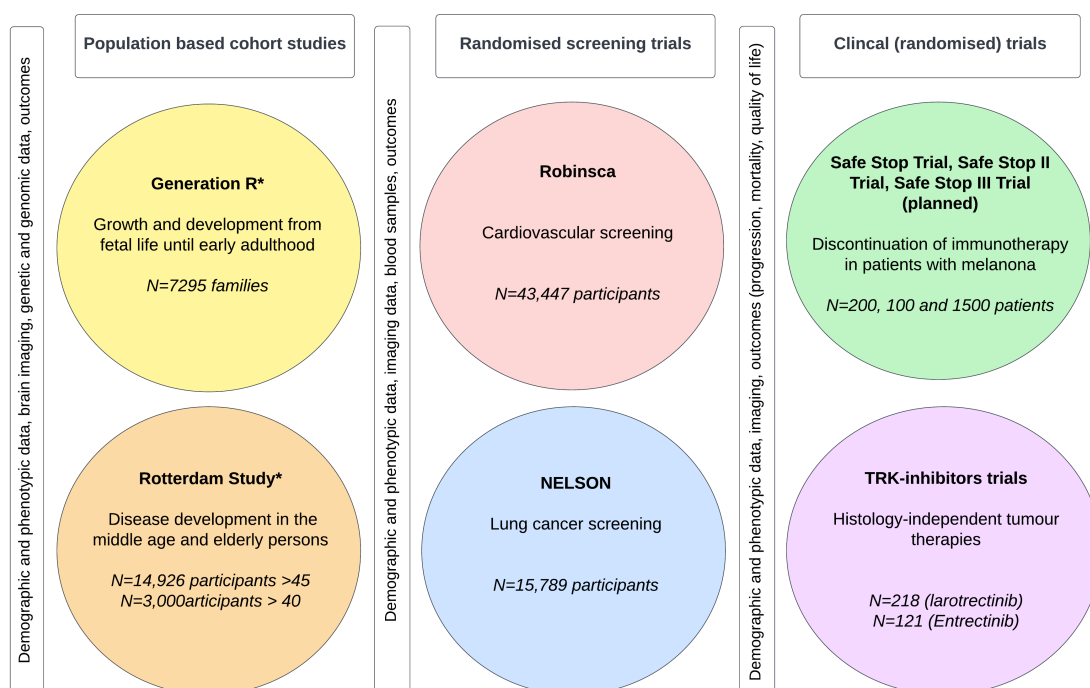
**Table 2: Projects selected for in-cash matching**

| Funder/Description budget sheet | Description/Name                                   | Status* | (Local) PI               | Start-end date (mm/yy) |
|---------------------------------|----------------------------------------------------|---------|--------------------------|------------------------|
| NWO-Vidi                        |                                                    | A       | Lansdorp-Vogelaar        | 09/21-08/26            |
| KWF                             | Dutch Cancer Society; NELSON-POP                   | A       | Uitterlinden/Stadhouders | 01/22-12/25            |
| Hartstichting                   | Dutch Heart Foundation; Dekkerbeurs                | A       | Kavousi                  | 06/22-05/26            |
| EU-Horizon Europe               | 4D-PICTURE                                         | A       | Rietjens                 | 09/22-09/27            |
| NWA-ORC                         | Dutch National Research Agenda; SENSE              | P       | Uitterlinden             | 03/23-08/30            |
| ZonMw                           | Dutch national funding organization for healthcare | P       | Kavousi                  | 03/23-02/27            |
| NWO                             | Dutch National Research Agenda                     | S       | Joint submission         |                        |
| EU-EIT Health                   | InEurHearth                                        | A       | Rutten                   | 01/22-12/24            |
| NWO-Vidi                        |                                                    | A       | Alfons                   | 12/19-12/24            |
| NWO-Vidi                        |                                                    | A       | de Bekker                | 09/21-09/26            |
| EU-ERC                          | ERC Consolidator                                   | P       | Uyl                      | 02/23-01-28            |
| EU Marie Curie                  | Network grant                                      | P       | Van Kippersluis          | 03/23-02/29            |

\*Status is Awarded (A) or Pipeline (P) or to be submitted (S)

We will use data directly available (Figure 1), so all WPs can **immediately start**. The Rotterdam Study will be used for refining individual predictions of **coronary artery disease risk** by combining traditional risk scores with polygenic risk score and coronary artery calcification measures. We have access to the **ROBINSca** trial which evaluates screening for cardiovascular diseases based on traditional risk scores and coronary artery calcification. We will use the **Generation R** study for the development of better **detection tools for ADHD**. Through involvement in multiple consortia we have access to additional cohorts for validation. To **personalise lung cancer screening** we will use RCT data from the **NELSON** trial to develop for risk based screening models, including personalisation of screening intervals. For **pharmacogenetic profiling** in the cardiovascular domain, we use data from the **Rotterdam study** to investigate the 'loss-of-function' risk alleles of CYP2C19 in users of clopidogrel because of stroke or myocardial infarction and published meta-analyses on clopidogrel effectiveness to model of a pharmacogenetic profile app on health and economic outcomes. For **personalisation of immunotherapy in melanoma**, we use data from the **Safe Stop I, II and (planned) III trials**. For evaluation of **histology-independent cancer therapies** we will use evidence from the trials on the TRK-Inhibitors for tumours with neurotrophic receptor tyrosine kinase (NTRK) gene fusions. Over time, the Flagship will evolve to other case-studies to cover the entire health journey from prevention to treatment in each of the disease areas.

**Figure 1: Data directly available to the Flagship NG- PM4all**



\*Generation R and the Rotterdam study are Core facilities of the Erasmus MC and are part of the Flagship 'A Lifecourse and Individual-based View on lifestyle to Enhance health (ALIVE)', PI's Dr. Luik, Kraal, and Voortman with whom we collaborate.

### 3.c Work plan

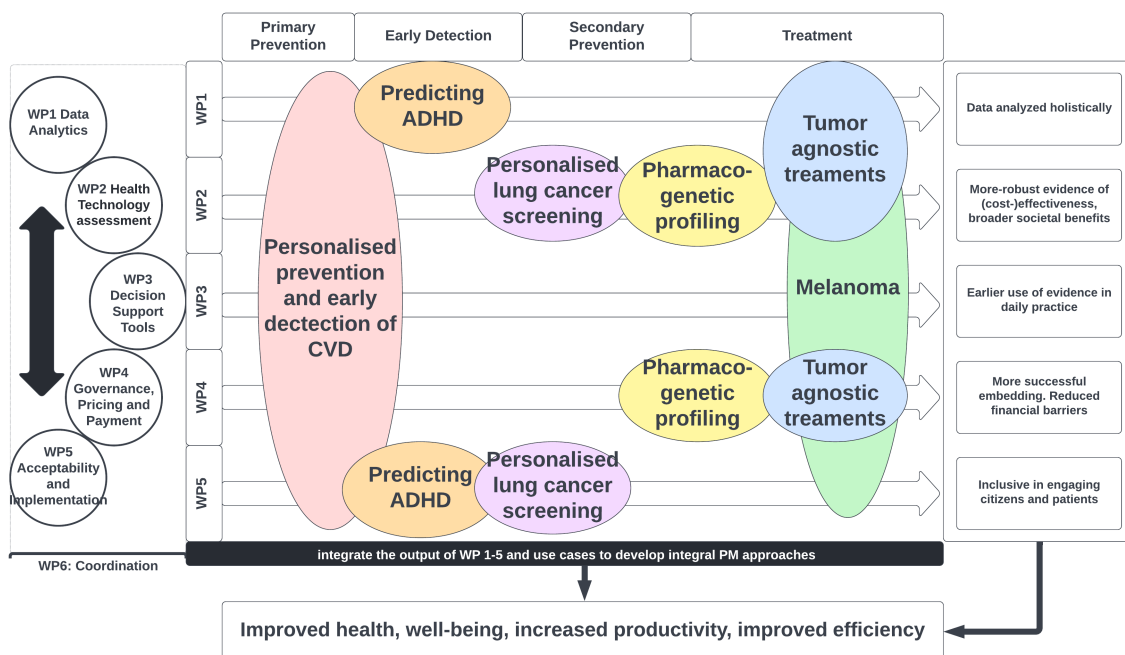
To realize the full potential of PM, we move beyond data, information and technology towards recognition of both people and technologies and their interdependencies within the organizational context. We take a **mixed-method approach** that incorporates both quantitative and qualitative methodologies. The flagship contains 5 work-packages, that address specific aspects of PM and do applied work on the case-studies. Integration in the 6<sup>th</sup> work package will result in PM approaches with the potential to reshape health journeys (Table 3, Figure 2). **The case studies ensure the link between the different scientific approaches to PM.**

**Table 3: Content of the WPs**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WP 1</b>   | <b>Data-analytics</b> (Fok, Klaveren)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Aim:          | Develop advanced analytic methodology to facilitate effective interventions for individuals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Tasks:        | Develop and improve methodology for:<br>1)Prediction of risk in <b>high-dimensional</b> data<br>2)Prediction of <b>heterogeneous intervention effects</b> in randomised studies<br>3)Personalised <b>causal inference</b> in observational data<br>4) <b>Personalised timing of interventions</b> , such as optimal screening intervals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Case studies: | Prediction of ADHD risk using multimodal, developmentally-sensitive (genetic) markers, prediction of CVD risk using multimodal data and subsequent optimisation of screening intervals, prediction of benefit and harm from immunotherapy for melanoma and for patients receiving histology-independent cancer treatments using randomised and non-randomised data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>WP 2</b>   | <b>Health Technology assessment</b> (Rutten, ten Haaf)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Aim:          | Evaluate the (cost-)effectiveness and full societal value of PM as compared to non-PM, using clinical decision-analytic and health economic modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Tasks:        | 1)Develop <b>open-access full disease-models</b> that cover the entire health journey, such that personalised health promotion, prevention and treatment can be assessed with the same cost-effectiveness model.<br>2)Develop methodologies to <b>account for greater uncertainties</b> in available evidence resulting from risk-stratification, for example comparing different methods to create synthetic control arms in case a comparator group is lacking.<br>3)Investigate methods of including <b>additional elements of value</b> beyond the quality-adjusted life year and their consequences for decision-making, for example the value of knowing and real-option value.<br>4)Contribute to implementable and responsible PM by considering distributional effects and costs across sub-groups and investigating the <b>ethical and legal</b> considerations of risk-stratification.<br>5)Develop an appropriate methodology to integrate all relevant aspects of a full HTA of PM, based on <b>multi-criteria decision analysis</b> (MCDA). |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case studies: | Evaluate pharmacogenetic profiling and evidence-based dosing guidelines in cardiology, oncology, and mental illness, personalised screening for lung cancer, risk screening followed by lifestyle and medical interventions for coronary artery disease, and personalised immunotherapy in melanoma, and tumour-agnostic treatments.                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>WP 3</b>   | <b>Decision support tools</b> (Kavousi, Chorus)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Aim:          | Develop transparent and easy-to-use decision support tools to facilitate responsible uptake of PM interventions, by health professionals and patients.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Tasks:        | <p>1) <b>Modelling the judgment</b> of medical professionals and patients regarding primary and secondary prevention, early detection and treatment. Next to objective factors, patient preferences are input for the professional judgment and vice versa.</p> <p>2) Specifically address <b>morally sensitive decision contexts</b> (like decisions on prevention and treatment etc.); take for example taboo trade off aversion, strategic/rational ignorance and moral satisficing (e.g., choosing the default).</p> <p>3) Studying (ways to avoid) the artificial alignment – ‘<b>echo chamber</b>’ – of preferences between medical professionals, between patients, and between medical professionals and patients.</p>           |
| Case studies: | Develop decision support tools for potential decisions regarding early detection of CVD and immunotherapy for melanoma treatment, that display recommendations for alternative decisions, including associated motivations and corresponding benefits and disadvantages to support trade-offs.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>WP 4</b>   | <b>Governance, Pricing, and Payment</b> (Uyl-de Groot, Uitterlinden)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Aim:          | Investigate pricing, payment and reimbursement models and healthcare systems and their governance that support the deployment of cost-effective PM at a larger scale                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Tasks         | <p>1) Investigate the most <b>optimal infrastructure and organisational embedding</b> of PM with respect to quality, efficiency, and access.</p> <p>2) Further develop and test <b>innovative pricing models</b> for PM that strike an optimal balance between improving equal access and stimulating innovation.</p> <p>3) Develop and apply flexible <b>budget impact models</b> that incorporate output from the full disease models and payment schemes to allow long-term financial planning.</p> <p>4) Develop new appropriate <b>reimbursement models/payment schemes</b> for expensive health technologies and assessing their long-term impact on the competition between companies operating in health technology markets.</p> |
| Case studies: | Personalised immunotherapy in melanoma, pharmacogenetic profiling in pharmacy, population screening strategies that include polygenic risk scores.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WP 5</b>   | <b>Acceptability and implementation</b> (Rietjens and Mouter)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Aim:          | Embed values and preferences of citizens, patients, and clinicians in the design of PM to ensure societal support and a smooth implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Tasks:        | <p>1)<b>Focus groups</b> with patients, clinicians and members of the general public to explore knowledge about personalised medicine, values and preferences about personal medicine, and barriers and facilitators.</p> <p>2)Conducting a <b>Participatory Value Evaluation</b> to identify shared values and preferences for PM</p> <p>3)Interviews with public, patients and clinicians about how they use PM in a real-life setting and what the <b>barriers and facilitators are regarding the use of PM</b></p> <p>4)Integrating decision support tools in patients' care paths using a promising <b>service design methodology, called MetroMapping</b>, in collaboration with designers.</p> |
| Case studies: | Study values and preferences to embed these in the PM instruments for immunotherapy in melanoma and CVD risk prediction as first cases. Integrate decision support tools for PM into patients' health journeys, starting with immunotherapy for melanoma.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>WP 6</b>   | <b>Integration and management</b> (Lingsma and Rutten)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Aim:          | Integrate the scientific and applied output of WP1-5 to develop integral PM approaches                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Tasks:        | <p>1)Integrate the methodological results of each WP in a <b>novel conceptual framework for PM</b>, addressing all steps from development of PM approaches to evaluation to implementation.</p> <p>2)<b>Link methods, models, tools, and recommendations</b> to guide the personalisation of interventions to the conceptual framework</p> <p>3)<b>Integrate the case-specific results</b> of each WP for each case-study in a <b>roadmap to PM</b> for that specific field, with its specific challenges and next steps.</p> <p>4)Coordinate management and convergence activities (section 4d)</p>                                                                                                  |
| Case studies: | All                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Figure 2: Workpackages and case-studies**

The personalised risk prediction and treatment models developed in WP1 will feed into WP2 for evaluation from a societal perspective and into WP3 for implementation in personalised decision tools and modelling of the decision-making process. WP4 will suggest adequate pricing and reimbursement models for the PM interventions and specify conditions for optimal implementation in clinical practice. The personalised decision tools from WP3 feed into WP5 for integration in health journeys and for critical appraisal by patients and public. The other way around, values and preferences of patients and public will be fed back into WP3 to be integrated in the decision support tools and the decision-making process, and into WP2 to be considered in the evaluation of PM approaches.

### 3.d Alignment of the program with national and international research agendas

#### *National*

NG-PM4all directly contributes to multiple research priorities of the **Dutch National Research Agenda**, which illustrates that PM is among the top priorities for research in the next decade, with with research questions like: “How can we personalise healthcare, for example by using biomarkers?” and “How will our knowledge of genetics play a role in analysing, screening for, and treating diseases?”

Our focus on personalised prevention aligns with the **National Prevention Agreement** and the **Dutch Coalition Agreement** putting emphasis on prevention. Recently, the joint academic medical centres have initiated a large application for a “**Groeifonds**”, focused on prevention and another “**Groeifonds**” on implementing tools for PM into various health care settings. The recent **WRR report “Choosing for sustainable care”** calls for urgent solutions to maintain affordability of healthcare. PM can be part of that by potentially improving effectiveness and accessibility.

### ***International***

At EU level we are aligned with the **International Consortium for Personalised Medicine (ICPerMed)** and the **European Partnership for PM** that is being set-up and will fund transnational consortia, networks, and collaborations in PM in future (European Partnership, 2022). NG-PM4all is aligned with various ongoing and upcoming research agendas of the European Union, such as the **European Mission on Cancer, Horizon Europe's 2021-2024 strategic plan for health**, which calls for health promotion and disease prevention throughout the life-course, and the **European '1+Million Genomes'** initiative, which acknowledges the relevance of genetic data in prevention and care.

## **4. Consortium (1992/2000 words)**

### **4.a Track record of the consortium members**

The **scientific quality** of the consortium is reflected by their current positions in the organisations, and personal grants and awards (among others 7xVENI, 6xVIDI, 4xERC starting and consolidator grant and 7xErasmus MC fellowship).

Their **ability to attract external funding** is illustrated by their coordination of and participation in European consortia (14 Horizon projects, 1 EIT project, 1 IMI project, and multiple NWO, ZonMW, NWA, KWF, Hartstichting and other national grants). Four applicants received **internal funding from the Convergence Open Mind** call for the projects: "Convergence for Individualizing TRreatment Using Statistical approaches", "Patient sc-AI-nce: a method for integrating patient experiences in text mining analysis", and "Improving the Health Journey of Heart Failure Patients".

Most partners **teach** in various bachelor, (research) master and post-graduate programs. Overall, the partners currently supervise 88 MSc, 158 PhD students and engage 89 postdocs in their research projects.

Applicants have built a great track record in academic research, educational activities, translation of research into clinical practice, and making an impact on society and their combined experience is essential to realise the full potential of personalised prevention and treatment.

The **EMC members** are well versed in evaluating risk-factors for the development of diseases, precision diagnosis, pharmacogenetic interventions and targeted treatments. The EMC partners from Public Health are experts in prevention and early detection of diseases and in person-centred care, and are involved in numerous intervention studies, clinical implementation studies and randomized clinical trials.

**EUR members** from the Erasmus School of Economics apply their expertise in data science to enhance the evaluation of the development, progression and treatment of diseases, as well as optimize treatment decisions and preventive measures while accounting for individual characteristics and preferences. The EUR partners from the Erasmus School of Health Policy & Management excel in broad health technology assessment, cost-effectiveness modelling, patient and stakeholder preference measurement, health care systems analysis, funding, and payment models, and translating the research findings to public policies.

Finally, the **members from TU Delft** are essential in designing decision support and citizen participation tools for PM that promote the uptake of interventions, also among the hard-to-reach populations and in building societal support for PM to accelerate adoption and access to PM.

#### **4.b Expertise of the consortium members in relation to the aims and objectives of the program**

##### ***Leads***

**Prof. dr. Hester Lingsma (EMC)** is professor of Medical Decision Making and Prognostic Modelling. She published applications of personalised treatment models as lead author in top journals including BMJ and Stroke. She leads WPs in two H2020 projects on personalised treatment (in brain injury and oncology) and in a national consortium on ischemic stroke. She is a member of the management committee of the Erasmus Initiative Smarter Choices for Better Health.

**Prof.dr. Maureen Rutten-van Mölken (EUR)** is professor of Economic Evaluation of Innovations for Health with expertise on health technology assessment and health economic modelling of the cost-effectiveness and full societal impact of PM interventions. She will bring-in the knowledge generated in the Horizon2020 HEcoPerMed project in which she has published on the net benefits of PM and innovative funding and payment models for (R&D of) PM. She issued guidelines for health economic modelling of PM. Coordinating the Horizon2020 project SELFIE, she published on the mechanism of successful implementation of person-centred integrated care programs for patients with multi-morbidity.

**Prof.dr.ir. Caspar Chorus (TUD)** is an international frontrunner in the field of choice behaviour modelling, with applications in domains as diverse as health, transport, environment, and marketing. In recent years, he has pioneered a novel way of decision support based on behavioural theories, which is rapidly gaining attention in the scientific field as well as in practice. Regarding decision support systems, Chorus is co-founder of TU Delft spin-off Councyl, which has relevant experience with providing decision support to hospitals.

##### ***Other WP leaders***

**Dr. David van Klaveren (EMC)** is assistant professor Medical Decision Making. He enabled personalised treatment decision making for coronary heart disease patients, by deriving the SYNTAX scores for selection of bypass surgery versus angioplasty, and the PRECISE-DAPT score for selection of dual antiplatelet therapy duration after stent implantation.

**Prof. dr. Dennis Fok (EUR)** is professor of Econometrics and Data Science and head of the Econometric Institute. He is at the forefront of methodological research in data science and econometrics and will contribute to the advancement of the methodology to estimate heterogeneous (treatment) effects.

**Dr. Kevin ten Haaf (EMC)** is an assistant professor in screening evaluation. He is involved in studies that evaluate personalised screening, such as the Horizon2020 funded 4-IN-THE-LUNG-RUN trial that evaluates different aspects of personalised lung cancer screening and he received a VENI grant to evaluate personalised breast, colorectal and lung cancer screening.

**Dr. Maryam Kavousi (EMC)** is associate professor of Epidemiology of Cardiometabolic disorders. She is the recipient of various national and international grants including a VENI, a Dutch UNESCO – L'Oréal fellowship "For Women in Science" and 3-month fellowship for NIAS-KNAW as well as a number of Horizon2020, ZonMw and *Hartstichting* grants. She is leading large international projects on cardiovascular risk prediction and the value of polygenic risk scores cardiovascular disease.

**Prof.dr. André Uitterlinden (EMC)** is professor of Complex Genetics at Erasmus MC. He is a board member of the Dutch Society of Human Genetics and PI of the Genome of Europe working group as part of the EU-sponsored 1 million Genomes Project, as well as member of the Dutch coordination group for Genome of the Netherlands for the Dutch ministry of Public Health.

**Prof.dr. Carin Uyl-de Groot (EUR)** is professor of Health Technology Assessment. As a member of the Dutch Committee on Expensive Cancer Drugs and as a member of the Dutch Health Council she is perfectly equipped to translate research results into clinical and policy recommendations.

**Dr. Judith Rietjens (EMC)** an associate professor of patient and family engagement in medical care and decision making. She is PI on an EU grant on medical decision making in advanced cancer patients and an EU grant on decision support tools in oncology, combining data sciences, design research, and medical and social sciences (4D PICTURE).

**Dr. Niek Mouter (TUD)** is associate professor of Infrastructure Policy Appraisal at TU Delft, with a background in economics and philosophy of law. He contributes to building a larger societal support base for PM that is aligned with the values and preferences of citizens, patients, and healthcare professionals using his tools for Participatory Value Evaluation which have generated positive acclaim in academia and practice alike.

#### *Other applicants*

| <b>EMC</b>                     | <b>Chair / Expertise</b>            |
|--------------------------------|-------------------------------------|
| Dr. Iris Lansdorp-Vogelaar     | Evaluation of screening             |
| Dr. Ida Korfage                | Quality of life and decision making |
| Prof. dr. Ron van Schaik       | Pharmacogenetics                    |
| Prof.dr. Pauline Jansen        | Developmental psychopathology       |
| Dr. Jeroen van Rooij           | Genetic risk profiling              |
| Prof.dr. Bruno Stricker        | Pharmacoepidemiology                |
| Dr. Astrid van der Veldt       | Medical oncology                    |
| Prof.dr. Natasja de Groot      | Translational electrophysiology     |
| Dr. Annemien van den Bosch     | Echocardiography                    |
| Prof.dr. Elisabeth van Rossum  | Obesity and biological stress       |
| Prof.dr. Manon Spaander        | Gastrointestinal oncology           |
| <b>EUR</b>                     |                                     |
| Prof.dr. Esther de Bekker-Grob | Economics and Preference Research   |
| Prof.dr. Anne Stiggelbout      | Medical Decision Making             |
| Prof.dr. Hans van Kippersluis  | Applied Economics                   |
| Dr. Andreas Alfons             | Statistics                          |
| Dr. Remy Spliet                | Operations Research                 |
| <b>TUD</b>                     |                                     |
| José Ignacio Hernández         | Participatory Value Evaluation      |
| Anna Marie de Ruijter          | Participatory Value Evaluation      |

#### 4.c Involvement of partners

During all phases of the research we will involve patients and public[6,7]. We already installed an initial **Patient and Public Involvement board (PPI)** (5 males, 1 female, all target group members of the interventions to be developed in this flagship), which **has provided feedback on our proposal**. We aim for a PPI board of 5-6 persons who will be active during the entire NG-PM4all study period. The panel will be diverse in terms of age, gender, health, and cultural background and will provide feedback on study progress, contribute to development of patient information and questionnaires, and will participate in meetings.

Besides engaging patients and the public, the consortium engages other stakeholders including professionals, payers, and policy makers. **The partners in NG-PM4all play a leading role in the field of PM and are very well connected to the external partners. Partners are key member of PM consortia and societies.** E.g.:

- Cohorts for Heart and Aging Research in Genomic Epidemiology (CHARGE),
- Global consortia for cardiovascular cohort studies (Cardiogram and C4D)
- Pregnancy and childhood epigenetics (PACE) consortium and Psychiatric Genetics Consortium (PGC)
- CISNET modelling consortium for screening
- The International Consortium for Personalised Medicine (ICPerMed)
- European Society of Cardiology (ESC) Council on Cardiovascular Genomics and Cardiovascular Risk Collaboration Unit
- Society for Medical Decision Making (SMDM)
- Dutch CardioVascular Alliance (DCVA)
- ‘Partnerschap Overgewicht Nederland’ (PON)
- International Society of Pharmacoeconomics and Outcomes Research (ISPOR)

We have close connections with both **policy makers** including the Rotterdam municipality and the National Institute for Public Health and the Environment (RIVM) and with **health care providers** including ‘Hart en Vaatcentrum Erasmus MC’, the CJG (Centres for youth healthcare), and several mental health care providers. We will further collaborate with 2 SMEs: Panton for design for health care and Council for decision support tooling, and the Institute for Medical Technology Assessment.

Finally, we have long-standing collaborations with **world-leading institutes in the field of PM** and with the other Dutch academic medical centers, e.g.:

- MRC Integrative Epidemiology Unit at the University of Bristol
- Predictive Analytics and Comparative Effectiveness center at Tufts Medical Center Boston
- Estonian Genome Center at Tartu University, Estonia
- Universidad de Zaragoza and Kolding Design School
- Choice modelling center Leeds
- CORIB Research Centre for Advanced Imaging and Core Laboratory at National University of Ireland Galway
- Health Economics Genomics England Clinical Interpretation Partnership

We will align the work within NG-PM4all with relevant **Academic Centres of Excellence** (ACE's) of the Erasmus MC in which partners play a central role; the ACE for Screening and Prevention, ACE Obesity Centre, ACE GESTURE (Gender Specific Translational research Union Rotterdam, Erasmus MC) and ACE Center for Quantitative Methods (CQM).

To ensure involvement from all these end-users right from the start we will organize a **kick-off event with different workshops** to discuss our research questions and approaches with them, followed by subsequent bi-annual workshops.

#### 4.d Governance, organization and collaboration strategy of the consortium

##### Organisational structure

- The **Steering Committee**, which includes the Flagship Lead (chair), Institute Leads, WP Leads, and scientific project manager, and meets bi-annually to decide on the overall strategy towards achieving project objectives, monitor overall progress and make contingency plans. It will also ensure strategic alignment of NG-PM4all with the goals of Convergence.
- The **Executive Committee**, which includes the Flagship leader, the two institute leads, and two post-doctoral scientific project managers who coordinate the collaboration across WPs, case studies and organisations. Administrative responsibilities will be delegated to the support staff of the Flagship lead. The executive committee meets bi-monthly and external funding will be a fixed agenda item.
- The **Consortium Assembly** including all applicants, chaired by a rotating WP leader. The assembly will meet every 6 months to discuss progress and interdisciplinary collaboration.
- The **Education Committee** which will be responsible for the Minor Program and opportunities for BSc and MSc students to engage in Flagship projects as part of their graduation thesis. This committee will comprise senior experts and early career researchers and postdocs involved in interdisciplinary teaching activities.
- The **Communication and Dissemination Committee** which organizes the bi-annual stakeholder workshops, seminar series, keynote lectures, disseminates findings to general audiences including social media outreach, and generates new routes towards impact for society. It is supported by the support staff.

##### Collaboration strategy

- Appointing a lead from one discipline and a co-lead from another discipline to each WP
- Employ postdocs in WP6 for the day-to-day integration of insights from various disciplines and the collaboration across WPs, case studies and institutions
- Supervising teams of the PhD students will include two or more different disciplines
- Organisation of interdisciplinary workshops on the case studies
- Secondments and visiting researchers

The main lead and two applicants (Prof. Lingsma, Prof. de Bekker and Prof. van Kippersluis) are members of the management committee of Smarter Choices for Better Health, and will contribute their **convergence lessons learned**, for example attention for cultural (e.g., publication culture) differences.

**Annexes**

**I Budget table**

**II Letters of intent and letters of commitment**

**III Letter of support from department head of the main lead**

**IV References**

## Budget formulier: Convergentie Flagship Call

|                                         |                                              |  |              |
|-----------------------------------------|----------------------------------------------|--|--------------|
| Flagship:                               | Next Genration Personalised Medicine for All |  |              |
| Flagship acronym:                       | NG-PP4all                                    |  |              |
| Naam Penvoerder:                        | H.F. Lingsma                                 |  |              |
| Startdatum en looptijd van het project: | 31-12-2023                                   |  |              |
| Totaal project budget:                  | €                                            |  | 3.999.906,00 |

### R&D Overview: kostensoorten per type onderzoek (hele project periode)

| 1.                             | Loonkosten:                 |                        |            |            |            |            |             |           | 2022    |         |
|--------------------------------|-----------------------------|------------------------|------------|------------|------------|------------|-------------|-----------|---------|---------|
|                                | Partner die de kosten maakt | Functie                | Aantal fte | Startdatum | Einddatum  | Inschaling | Maandbedrag | Kosten    | in cash | in kind |
|                                | Onderzoeksorganisaties      |                        |            |            |            |            |             |           |         |         |
|                                | Erasmus MC                  | PhD (OIO)              | 1,00       | 1-1-2023   | 31-12-2026 | UMC - OIO  | € 3.259     | € 234.677 |         |         |
|                                | Erasmus MC                  | PhD (Arts-onderzoeker) | 1,00       | 1-1-2023   | 31-12-2026 | UMC - 10   | € 3.861     | € 278.006 |         |         |
|                                | Erasmus MC                  | Postdoc                | 0,50       | 1-1-2023   | 31-12-2027 | UMC - 11   | € 4.868     | € 219.081 |         |         |
|                                | Erasmus MC                  | Support staff          | 0,32       | 1-1-2023   | 31-12-2027 | UMC - 7    | € 3.817     | € 109.930 |         |         |
|                                | EUR                         | PhD                    | 2,00       | 1-1-2023   | 31-12-2026 | UNI - p    | € 3.793     | € 546.225 |         |         |
|                                | EUR                         | Postdoc                | 0,50       | 1-1-2023   | 31-12-2026 | UNI - 11   | € 4.729     | € 170.226 |         |         |
|                                | TU Delft                    | PhD                    | 1,50       | 1-1-2023   | 31-12-2026 | UNI - p    | € 3.122     | € 337.176 |         |         |
|                                |                             |                        |            |            |            |            |             | € -       |         |         |
|                                |                             |                        |            |            |            |            |             | € -       |         |         |
|                                | Ondernemingen               |                        |            |            |            |            |             |           |         |         |
|                                | Onderneming 1               |                        |            |            |            |            | € -         | € -       |         |         |
|                                | Onderneming 2               |                        |            |            |            |            | € -         | € -       |         |         |
| Onderneming 3                  |                             |                        |            |            |            | € -        | € -         |           |         |         |
| Onderneming 4                  |                             |                        |            |            |            | € -        | € -         |           |         |         |
| Overige (private) partners     |                             |                        |            |            |            |            |             |           |         |         |
| Erasmus MC - NWO-Vidi          | PhD (OIO)                   | 1,00                   | 1-1-2023   | 31-12-2024 | UMC - OIO  | € 3.329    | € 119.858   |           |         |         |
| Erasmus MC - KWF               | PhD (OIO)                   | 1,00                   | 1-1-2023   | 31-12-2024 | UMC - OIO  | € 3.329    | € 119.858   |           |         |         |
| Erasmus MC - Hartstichting     | PhD (OIO)                   | 0,50                   | 1-1-2023   | 31-12-2024 | UMC - OIO  | € 2.970    | € 53.453    |           |         |         |
| Erasmus MC - EU-Horizon Europe | PhD (OIO)                   | 1,00                   | 1-1-2023   | 31-12-2024 | UMC - OIO  | € 2.970    | € 106.906   |           |         |         |
| Erasmus MC - EU-Horizon Europe | Postdoc                     | 0,50                   | 1-1-2023   | 31-12-2024 | UMC - 11   | € 4.506    | € 81.106    |           |         |         |
| Erasmus MC - EU-Horizon Europe | Support staff               | 0,25                   | 1-1-2023   | 31-12-2024 | UMC - 7    | € 3.817    | € 34.353    |           |         |         |
| Erasmus MC - NWA-ORC           | PhD (OIO)                   | 1,00                   | 1-1-2026   | 31-12-2027 | UMC - OIO  | € 3.259    | € 117.339   |           |         |         |
| Erasmus MC - NWA-ORC           | Postdoc                     | 0,50                   | 1-1-2026   | 31-12-2027 | UMC - 11   | € 7.293    | € 131.268   |           |         |         |
| Erasmus MC - NWA-ORC           | Support staff               | 0,25                   | 1-1-2025   | 31-12-2027 | UMC - 7    | € 3.817    | € 51.530    |           |         |         |
| Erasmus MC - ZonMw             | PhD (OIO)                   | 1,00                   | 1-1-2025   | 31-12-2026 | UMC - OIO  | € 3.259    | € 117.339   |           |         |         |
| Erasmus MC - NWO               | PhD (OIO)                   | 1,00                   | 1-1-2025   | 31-12-2026 | UMC - OIO  | € 3.259    | € 117.339   |           |         |         |
| EUR - EU-EIT Health            | PhD                         | 1,00                   | 1-1-2023   | 31-12-2024 | UNI - p    | € 3.793    | € 136.548   |           |         |         |
| EUR - NWO-Vidi                 | PhD                         | 1,00                   | 1-1-2023   | 31-12-2024 | UNI - p    | € 3.793    | € 136.548   |           |         |         |
| EUR - NWO-Vidi                 | Postdoc                     | 0,50                   | 1-1-2023   | 31-12-2024 | UNI - 11   | € 4.729    | € 85.113    |           |         |         |
| EUR - EU-ERC                   | PhD                         | 1,00                   | 1-1-2026   | 31-12-2027 | UNI - p    | € 3.793    | € 136.548   |           |         |         |
| EUR - EU-ERC                   | Postdoc                     | 0,50                   | 1-1-2026   | 31-12-2027 | UNI - 11   | € 4.729    | € 85.113    |           |         |         |
| EUR - EU-Marie Curie           | PhD                         | 1,00                   | 1-1-2025   | 31-12-2026 | UNI - p    | € 3.793    | € 136.548   |           |         |         |
| TU Delft - NWO                 | PhD                         | 1,50                   | 1-1-2025   | 31-12-2026 | UNI - p    | € 3.122    | € 168.588   |           |         |         |
|                                |                             |                        |            |            |            |            |             |           |         |         |
|                                |                             |                        |            |            |            |            |             |           |         |         |
| Naam 4                         |                             |                        |            |            |            |            | € -         | € -       |         |         |
|                                |                             |                        |            |            |            | Totaal     | € 3.830.674 |           |         |         |

[illegible]

3.

| Kosten van gebruik van machines en apparatuur |                      |  |  |                           |                   |            | I 2.022 |         |
|-----------------------------------------------|----------------------|--|--|---------------------------|-------------------|------------|---------|---------|
| Partner die de kosten maakt                   | Machine / apparatuur |  |  | Prijs per gebruikseenheid | Gebruiks-eenheden | Kosten     | in cash | in kind |
| <b>Onderzoeksorganisaties</b>                 |                      |  |  |                           |                   |            |         |         |
| Organisatie 1                                 |                      |  |  |                           |                   |            |         |         |
| Organisatie 2                                 |                      |  |  |                           |                   |            |         |         |
| Organisatie 3                                 |                      |  |  |                           |                   |            |         |         |
| Organisatie 4                                 |                      |  |  |                           |                   |            |         |         |
| <b>Ondernemingen</b>                          |                      |  |  |                           |                   |            |         |         |
| Onderneming 1                                 |                      |  |  |                           |                   |            |         |         |
| Onderneming 2                                 |                      |  |  |                           |                   |            |         |         |
| Onderneming 3                                 |                      |  |  |                           |                   |            |         |         |
| Onderneming 4                                 |                      |  |  |                           |                   |            |         |         |
| <b>Overige (private) partners</b>             |                      |  |  |                           |                   |            |         |         |
| Naam 1                                        |                      |  |  |                           |                   |            |         |         |
| Naam 2                                        |                      |  |  |                           |                   |            |         |         |
| Naam 3                                        |                      |  |  |                           |                   |            |         |         |
| Naam 4                                        |                      |  |  |                           |                   |            |         |         |
| <b>Totaal</b>                                 |                      |  |  |                           |                   | <b>I -</b> |         |         |

[illegible]

|    |                                     |                                 |        |         |           |         |   |
|----|-------------------------------------|---------------------------------|--------|---------|-----------|---------|---|
| 4. | Aan derden verschuldigde kosten     |                                 |        |         | 2.022     |         |   |
|    | Partner die de kosten maakt         | Naam derde, omschrijving kosten |        | Kosten  | in cash   | in kind |   |
|    | <b>Onderzoeksorganisaties</b>       |                                 |        |         |           |         |   |
|    | Organisatie 1                       |                                 |        |         |           |         |   |
|    | Organisatie 2                       |                                 |        |         |           |         |   |
|    | Organisatie 3                       |                                 |        |         |           |         |   |
|    | Organisatie 4                       |                                 |        |         |           |         |   |
|    | <b>Ondernemingen</b>                |                                 |        |         |           |         |   |
|    | Onderneming 1                       |                                 |        |         |           |         |   |
|    | Onderneming 2                       |                                 |        |         |           |         |   |
|    | Onderneming 3                       |                                 |        |         |           |         |   |
|    | Onderneming 4                       |                                 |        |         |           |         |   |
|    | <b>Overige (private) partners</b>   |                                 |        |         |           |         |   |
|    | Naam 1                              |                                 |        |         |           |         |   |
|    | Naam 2                              |                                 |        |         |           |         |   |
|    | Naam 3                              |                                 |        |         |           |         |   |
|    | Naam 4                              |                                 |        |         |           |         |   |
|    |                                     |                                 | Totaal | -       |           |         |   |
| 5. | Publicatie, reis- en verblijfkosten |                                 |        |         | 2.022     |         |   |
|    | Partner die de kosten maakt         | Naam derde, omschrijving kosten |        | Kosten  | in cash   | in kind |   |
|    | <b>Onderzoeksorganisaties</b>       |                                 |        |         |           |         |   |
|    | Erasmus MC                          | Publication costs               | 38%    | 15.385  |           |         |   |
|    | EUR                                 | Publication costs               | 38%    | 15.385  |           |         |   |
|    | TU Delft                            | Publication costs               | 23%    | 9.231   |           |         |   |
|    | Erasmus MC                          | Travel and congress costs       | 38%    | 10.215  |           |         |   |
|    | EUR                                 | Travel and congress costs       | 38%    | 10.115  |           |         |   |
|    | TU Delft                            | Travel and congress costs       | 23%    | 6.269   |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    | Organisatie 4                       |                                 |        |         |           |         |   |
|    | <b>Ondernemingen</b>                |                                 |        |         |           |         |   |
|    | Onderneming 1                       |                                 |        |         |           |         |   |
|    | Onderneming 2                       |                                 |        |         |           |         |   |
|    | Onderneming 3                       |                                 |        |         |           |         |   |
|    | Onderneming 4                       |                                 |        |         |           |         |   |
|    | <b>Overige (private) partners</b>   |                                 |        |         |           |         |   |
|    | Erasmus MC - NVA-ORC                | Publication costs               |        | 5.128   |           |         |   |
|    | Erasmus MC - ZonMw                  | Publication costs               |        | 5.128   |           |         |   |
|    | Erasmus MC - NWO                    | Publication costs               |        | 5.128   |           |         |   |
|    | EUR - EU-ERC                        | Publication costs               |        | 7.693   |           |         |   |
|    | EUR - EU-Marie Curie                | Publication costs               |        | 7.693   |           |         |   |
|    | TU Delft - NWO                      | Publication costs               |        | 9.231   |           |         |   |
|    | Erasmus MC - EU-Horizon Europe      | Travel and congress costs       |        | 1.923   |           |         |   |
|    | Erasmus MC - NWO-Vidi               | Travel and congress costs       |        | 1.923   |           |         |   |
|    | Erasmus MC - ZonMw                  | Travel and congress costs       |        | 2.885   |           |         |   |
|    | Erasmus MC - NWO                    | Travel and congress costs       |        | 2.885   |           |         |   |
|    | EUR - NWO-Vidi                      | Travel and congress costs       |        | 1.923   |           |         |   |
|    | EUR - EU-EIT Health                 | Travel and congress costs       |        | 1.923   |           |         |   |
|    | EUR - EU-ERC                        | Travel and congress costs       |        | 2.885   |           |         |   |
|    | EUR - EU-Marie Curie                | Travel and congress costs       |        | 2.885   |           |         |   |
|    | TU Delft - NWO                      | Travel and congress costs       |        | 6.269   |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    |                                     |                                 |        |         |           |         |   |
|    | Naam 3                              |                                 |        |         |           |         |   |
|    | Naam 4                              |                                 |        |         |           |         |   |
|    |                                     |                                 | Totaal | 132.100 |           |         |   |
| 6. | Totale kosten                       |                                 |        |         | 4.000.774 | -       | - |



# FINANCIERING - Bijdragen per partner per jaar

| in Euro                          | 2.022   |         | 2.023   |         | 2.024   |         | 2.025   |         | 2.026     |         | 2.027   |         | Totaal    |         | Totaal per partner |  |
|----------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|-----------|---------|---------|---------|-----------|---------|--------------------|--|
|                                  | in cash | in kind | in cash | in kind | in cash | in kind | in cash | in kind | in cash   | in kind | in cash | in kind | in cash   | in kind |                    |  |
| Onderzoekorganisaties            |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
| 1 Erasmus MC                     |         |         | 182.650 |         | 198.836 |         | 213.789 |         | 224.523   |         | 85.495  |         | 905.294   | -       | 905.294            |  |
| 2 EUR                            |         |         | 181.036 |         | 181.036 |         | 186.164 |         | 186.164   |         | 7.551   |         | 741.951   | -       | 741.951            |  |
| 3 TU Delft                       |         |         | 85.448  |         | 85.448  |         | 88.525  |         | 88.525    |         | 4.731   |         | 352.676   | -       | 352.676            |  |
| 4 Organisatie 4                  |         |         | -       |         | -       |         | -       |         | -         |         | -       |         | -         | -       | -                  |  |
| 5 Organisatie 5                  |         |         | -       |         | -       |         | -       |         | -         |         | -       |         | -         | -       | -                  |  |
| 6 Organisatie 6                  |         |         | -       |         | -       |         | -       |         | -         |         | -       |         | -         | -       | -                  |  |
| Totale bijdrage onderzoeksorga   | -       | -       | 449.134 | -       | 465.320 | -       | 488.478 | -       | 499.212   | -       | 97.778  | -       | 1.999.921 | -       | 1.999.921          |  |
| Ondernemingen                    |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
| 1 Onderneming 1                  |         |         |         |         |         |         |         |         |           |         |         |         | -         | -       | -                  |  |
| 2 Onderneming 2                  |         |         |         |         |         |         |         |         |           |         |         |         | -         | -       | -                  |  |
| 3 Onderneming 3                  |         |         |         |         |         |         |         |         |           |         |         |         | -         | -       | -                  |  |
| 4 Onderneming 4                  |         |         |         |         |         |         |         |         |           |         |         |         | -         | -       | -                  |  |
| Totale bijdrage ondernemingen    | -       | -       | -       | -       | -       | -       | -       | -       | -         | -       | -       | -       | -         | -       | -                  |  |
| Overige (private) partners       |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
| 1 Erasmus MC - NwA-ORC           |         |         | -       |         | -       |         | 66.061  |         | 119.602   |         | 119.602 |         | 305.264   | -       | 305.264            |  |
| 2 Erasmus MC - ZonMw             |         |         | -       |         | -       |         | 59.631  |         | 64.759    |         | 962     |         | 125.351   | -       | 125.351            |  |
| 3 Erasmus MC - NwO-Vidi          |         |         | 59.020  |         | 62.762  |         | -       |         | -         |         | -       |         | 121.781   | -       | 121.781            |  |
| 5 Erasmus MC - Kwf               |         |         | 58.058  |         | 61.800  |         | -       |         | -         |         | -       |         | 119.858   | -       | 119.858            |  |
| 6 Erasmus MC - Hartstichting     |         |         | 24.424  |         | 29.029  |         | -       |         | -         |         | -       |         | 53.453    | -       | 53.453             |  |
| 7 Erasmus MC - EU-Horizon Europe |         |         | 106.382 |         | 117.906 |         | -       |         | -         |         | -       |         | 224.288   | -       | 224.288            |  |
| 8 Erasmus MC - NwO               |         |         | -       |         | -       |         | 59.631  |         | 59.631    |         | 6.090   |         | 125.351   | -       | 125.351            |  |
| 9 EUR - EU-ERC                   |         |         | -       |         | -       |         | -       |         | 119.965   |         | 112.273 |         | 232.238   | -       | 232.238            |  |
| 10 EUR - NwO-Vidi                |         |         | 111.792 |         | 111.792 |         | -       |         | -         |         | -       |         | 223.584   | -       | 223.584            |  |
| 11 EUR - EU-EIT Health           |         |         | 69.236  |         | 69.236  |         | -       |         | -         |         | -       |         | 138.471   | -       | 138.471            |  |
| 12 EUR - EU-Marie Curie          |         |         | -       |         | -       |         | 69.236  |         | 69.236    |         | 8.654   |         | 147.125   | -       | 147.125            |  |
| 13 TU Delft - NwO                |         |         | -       |         | -       |         | 86.994  |         | 91.610    |         | 4.616   |         | 183.219   | -       | 183.219            |  |
| 14 Flagship                      |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
| ... Naam 4                       |         |         |         |         |         |         |         |         |           |         |         |         | -         | -       | -                  |  |
| Totale overige bijdrage          | -       | -       | 428.911 | -       | 452.524 | -       | 341.552 | -       | 524.802   | -       | 252.195 | -       | 1.999.984 | -       | 1.999.984          |  |
|                                  |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
|                                  | in cash | in kind | in cash | in kind | in cash | in kind | in cash | in kind | in cash   | in kind | in cash | in kind | Totaal    |         |                    |  |
| Totaal budget per jaar           | -       | -       | 878.045 | -       | 917.844 | -       | 830.030 | -       | 1.024.014 | -       | 349.973 | -       | 3.999.905 |         |                    |  |
|                                  |         |         |         |         |         |         |         |         |           |         |         |         |           |         |                    |  |
|                                  |         |         |         |         |         |         |         |         |           |         |         |         |           |         | Funding Flagship   |  |
| Totaal budget                    | -       | -       | 878.045 | -       | 917.844 | -       | 830.030 | -       | 1.024.014 | -       | 349.973 | -       | 3.999.905 |         |                    |  |

|                           |           |
|---------------------------|-----------|
| Erasmus MC / Convergentie | 1905.294  |
| EUR / Convergentie        | 1741.951  |
| TU Delft / Convergentie   | 1352.676  |
| Totaal Convergentie       | 1.999.921 |

|                       |            |
|-----------------------|------------|
| Erasmus MC / Matching | 11.075.347 |
| EUR / Matching        | 1741.418   |
| TU Delft / Matching   | 183.219    |
| Totaal Matching       | 1.999.984  |



Prof.dr. A. Burdorf  
Erasmus Medical Center  
Department of Public Health  
P.O. Box 2040  
3000 CA Rotterdam

10 March 2022

***Letter of in intent/support for project 'Next generation personalised medicine for all'***

Dear Prof. Burdorf,

I, Jeanine Roeters van Lennep, in my capacity of Executive Committee (*dagelijks bestuur*) of *het Hart Vaatcentrum* at Erasmus MC hereby confirm that *het Erasmus MC Hart en Vaat Instituut* supports the project ***Next generation personalised medicine for all*** as applied for by the main applicant, Hester Lingsma, Professor of Medical Decision Making at the Department of Public Health, Erasmus MC.

*Het Erasmus MC Hart en Vaat Instituut* supports the project and intends to contribute knowledge and expertise, further detailed in the project proposal.

Yours sincerely,

Signature:

A handwritten signature in blue ink, reading "Roeters van Lennep", with a long horizontal line extending from the end of the signature.

Name: Jeanine Roeters van Lennep  
Position: Internist, Vascular Medicine  
Date: 10 March 2022

To: Prof.dr. Maureen Rutten-van Mölken and Prof.dr. Hester Lingsma  
Erasmus University Rotterdam  
Rotterdam School of Health Policy and Management  
Department of HTA  
P.O. Box 1738  
3000 DR Rotterdam  
The Netherlands

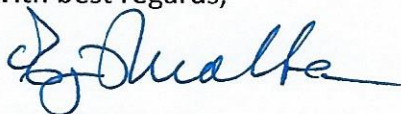
Dear Profs Rutten-van Mölken and Lingsma

I am writing to express the International Consortium for Personalised Medicine's endorsement of the Flagship "Next Generation Personalized Medicine for all", funded by the Convergence Health and Technology of the Erasmus Medical Center, the Erasmus University in Rotterdam, and the Technical University of Delft in the Netherlands.

ICPerMed considers this initiative to be of high value to the field of personalised medicine and can therefore strongly support that the project is funded. The project will undoubtedly contribute to the reasonable and fair implementation of personalized medicine approaches into the health systems for the benefit of patients, citizens, and society.

When relevant, the Flagship can exchange knowledge on personalized medicine with initiatives under the ICPerMed family umbrella and participate in ICPerMed events.

With best regards,



Ejner Moltzen  
Innovation Fund Denmark  
Chair of ICPerMed

*ICPerMed is a member state driven activity which provides a platform to initiate and support communication and exchange on personalised medicine research, funding, and implementation. The high level of participation from all over Europe and beyond enables ICPerMed to efficiently map the scientific and political landscape.*

*ICPerMed works in close collaboration with the European Research Area Network for Personalised Medicine (ERA PerMed), several Coordination and Support Actions (CSAs) funded by the European Commission, as well as with an increasing number of associated and related initiatives, research infrastructures and capacities. This so called ICPerMed 'Family' has an important role in supporting the research and implementation of personalised medicine in Europe and beyond.*



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Prof.dr. A. Burdorf  
Erasmus Medical Center  
Department of Public Health  
P.O. Box 2040  
3000 CA Rotterdam

11 March 2022

***Letter of in intent/support for project 'Next generation personalised medicine for all'***

Dear Prof. Burdorf,

I, Bianca Brundel, in my capacity of founder and director of AFIP foundation (<https://afiponline.org/>) hereby confirm that AFIP foundation supports the project ***Next generation personalised medicine for all*** as applied for by the main applicant, Hester Lingsma, Professor of Medical Decision Making at the Department of Public Health, Erasmus MC.

AFIP foundation supports the project and intends to contribute knowledge and expertise, further detailed in the project proposal.

Yours sincerely,

Signature:  
Name: Prof. Dr. B. Brundel  
Position: Molecular Biologist  
Date: 11 March 2022



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**Direct dial** +3110 703 84 69  
**Internal postal address** NA-2401  
**E-mail** a.burdorf@erasmusmc.nl  
**Our reference** LBMGZJS  
**Date** 14-03-2022

To whom it may concern,

In my capacity of head of the department of Public Health of the Erasmus MC, I hereby want to express my support for submission of the flagship proposal 'Next Generation Personalised Medicine for all' (NG-PM4all) led by Prof.dr. Hester Lingsma from my department, and co-led by Prof.dr. Maureen Rutten-van Mölken from the Erasmus School of Health Policy and Management of the Erasmus University and Prof.dr. Caspar Chorus of Engineering Systems and Services of the TU Delft.

**Postal address**  
P.O. box 2040  
3000 CA Rotterdam, NL

**Visiting address**  
Dr. Molewaterplein 40,  
3015 GD Rotterdam, NL

The aim of the flagship is to integrate several data sources, perspectives, and approaches to develop and improve personalized prevention and treatment strategies, to evaluate their impact on individuals and society, and to develop solutions and strategies for successful embedding into routine practice.

I am convinced this Flagship will contribute to the aims of the Health and Technology Convergence:

- Life-long health and well-being, from pre-birth to end of life
- Socio-economic equality in health(care)
- Prevention and early diagnostics to proactively maintain health
- Individual-tailored precise medical treatment for all

More specifically it perfectly aligns with the aim of the theme 'Improving Health Journeys', which is to promote individual health and well-being, develop and implement effective prevention strategies, early disease detection, and personalized treatment and support from a life course perspective.

Therefore, I support this application and hope it will be successful.



prof. dr. ir. A. Burdorf  
Head of Department Public Health  
Theme Health Sciences  
Erasmus MC

Date

March 22, 2022

Subject

Letter of intent

Our reference

SW/20222203/mr

Page

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Visiting address

ESHPM  
Burgemeester Oudlaan 50  
Bayle Building  
J8.33

Postal address

PO Box 1738  
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The Netherlands

T +31 10 408 86 96

E m.rutten@eshpm.eur.nl

W www.eur.nl/eshpm

**LETTER OF INTENT**

for the

**Flagship/Project**

**Next Generation Personalised Medicine for All (NG-PM4all)**

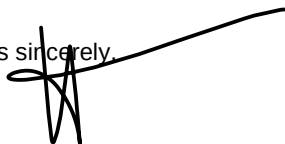
Dear professor Hester Lingsma (Erasmus MC),

I, prof. Werner Brouwer, in my capacity of pro Dean at Erasmus School of Health Policy & Management (ESHPM) hereby confirm that ESHPM intends to contribute to the Flagship *Next Generation Personalised Medicine for All (NG-PM4all)*, as applied for by the main applicant, professor Lingsma.

ESHPM intends to contribute to this flagship program:

- with the awarded projects Horizon2020 HEcoPerMed, Horizon2020 T2EVOLVE, ZonMW Precision, EIT Health InEurHeart, ZIN Melanoma model, personalised diagnosis and treatment of early rheumatoid arthritis, country-adaptation CEA-model for Spinal Muscle Atrophy, and the VIDI on preference measurement of Esther de Bekker Grob.
- if awarded, with Horizon Europe PREIS, and the ERC Consolidator Grant of Carin Uyl-de Groot.

Yours sincerely,



Name: prof. W. Brouwer

Position: pro Dean

Date: 22 March 2022

## References

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## Selection of relevant publications of the consortium

### Data-analytics and causal inference

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5. COVID outcome prediction in the emergency department (COPE): using retrospective Dutch hospital data to develop simple and valid models for predicting mortality and need for intensive care unit admission in patients who present at the emergency department with suspected COVID-19.
6. **van Klaveren D**, Rekkas A, Alsma J, Verdonschot RJCG, Koning DTJJ, Kamps MJA, Dormans T, Stassen R, Weijer S, Arnold KS, Tomlow B, de Geus HRH, van Bruchem-Visser RL, Miedema JR, Verbon A, van Nood E, Kent DM, Schuit SCE, **Lingsma H**. *BMJ Open*. 2021 Sep 16;11(9):e051468. doi: 10.1136/bmjopen-2021-051468.

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